TeachBack

Safety-weighted, component-level scoring of patients' medication teach-back

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LLM Applications course ▪ Bar-Ilan University ▪ Project proposal

Slide 1: Motivating use case

Use case background

- *Teach-back*: after explaining a medication, the nurse or pharmacist asks the patient to repeat the instructions in their own words, and checks what was understood.
- Standard practice at discharge and in pharmacies.

Why the problem is important

- Misunderstood instructions (wrong dose, stopping antibiotics early, mixing with alcohol) cause preventable harm at home.
- Teach-back improves recall and comprehension, but only if the clinician spots the gap.

Why it is challenging

- **Free text**: informal, partial, out of order, many valid paraphrases ("twice a day" = "every 12 hours").
- **Asymmetric risk**: one wrong dose outweighs five correct details; text similarity can't express that.
- **Costly**: 21.5% of 144 patients right on the first try; ≈23% needed 4–5 rounds.
- **No data**: no public set of instructions + restatements + labels.

How the problem is solved today

- By hand, by the clinician, with no record of what was missed.
- Research measures comprehension with questionnaires or multiple-choice exams, not the patient's own words.

Sources: PMC12888902 (cardiac teach-back QI study); ContextCare protocol NCT07890532; Bazaldua, BUMC Health Literacy Conference 2011.

Slide 2: Project task description

Formal problem statement

Given a medication instruction split into atomic components $c_1..c_n$ (each with category, weight $\tilde{w}_i$, safety-critical flag s_i) and a patient's free-text restatement R, learn $f(c_i, R) \to y_i \in \{$**D**emonstrated, **P**artial, **O**mitted, **C**ontradicted$\}$. A fixed rule then gives the score:

$$S = 100 \cdot \max\left(0, \sum_i \tilde{w}_i \, v(y_i)\right), \quad v = \{1,\ 0.5,\ 0,\ -0.5\}$$

Safety caps: a critical component **C** $\Rightarrow S \leq 40$; a critical component **O** or **P** $\Rightarrow S \leq 84$, and it is always listed as *must check*.

Review category: $\geq$85 no flagged misunderstanding ▪ 50–84 clarification recommended ▪ $<$50 substantial

Inputs and outputs

In	Components, e.g. "one 500 mg capsule per dose" ▪ dose ▪ critical
In	Restatement, e.g. "One pill three times a day, I stop when I feel better"
Out	Status + evidence span per component: dose = **D** ("One pill"), duration = **C**
Out	Score 57 ▪ clarification recommended ▪ must check: allergy warning (critical, omitted) ▪ revisit: duration

Project novelty

- Scores the *patient's own words* per instruction component, not a questionnaire, exam or the clinician's behaviour.
- Explicit, harm-weighted rule with two safety caps: a dangerous error never looks "mostly fine", and a missing critical item never passes as understood.
- Robustness test: equal scores for equal meaning, whatever the wording.

Slide 3: Models and methods

Processing flow / pipeline



Models / techniques in each step

2–3	GPT-class LLM, attribute-based generation (chosen status + style per component)
4	Verifier LLM from a different family + human audit sample
5	Compared: **off-the-shelf** E0 ROUGE/BERTScore + regression ▪ E1 LLM judge, zero/few-shot ▪ E2 zero-shot NLI (DeBERTa-v3 MNLI); **fine-tuned** E3a DeBERTa-v3 cross-encoder (main) ▪ E3b ModernBERT ▪ E3c direct score regression (ablation)
6	Deterministic formula, no learning

Adjustments and fine-tuning

- Input `[category] statement [SEP] restatement`, 4-class head; E3a starts from an NLI checkpoint.
- Class-weighted loss (**P** and **C** are rarer); early stopping on validation macro-F1.
- Sweep learning rate and epochs; 3 seeds each.
- E1: rubric + few-shot examples in the prompt, JSON output.
- Optional: LoRA-tuned 1–4B LLM.

Slide 4: Data specification and generation

Training and evaluation data requirements

- (component, restatement, status) triples with a known status.
- Many styles: terse, rambling, informal, non-native, low literacy.
- Hard cases: subtle contradictions and valid paraphrases.
- Test: unseen instructions, held-out drug classes, $\geq$300 restatements with a contradicted critical component.

Dataset

- No existing labelled dataset $\rightarrow$ synthetic, from DailyMed / openFDA drug labels.
- $\approx$400 instructions $\times$ 8 restatements $\approx$ 3,200 $\rightarrow$ $\approx$16–19k pairs.
- Split by source instruction (no leakage); held-out drug classes as a 2nd test.

Labeling

- Labels come from the generation request (status chosen per component).
- Verified by a second LLM; mismatches dropped; 150–200 pairs checked by hand.
- Challenge set: $\approx$150 role-played restatements labelled by two team members.

Synthetic data: what and how

- LLM rewrites label text into patient instructions as JSON components.
- Restatements generated from (status per component, style).
- Operators: paraphrases from an equivalence table; 50 mg for 500; "two pills once" for "one pill twice"; reversed alcohol rule; partials; omissions.
- Unclear equivalences $\rightarrow$ adjudicated, not used as exact labels.

Slide 5: Metrics and KPIs

How we measure the results

Primary: **component macro-F1** (quality) and **dangerous-miss rate** (safety, target ≈0). Each hypothesis is one comparison: fine-tuned vs off-the-shelf (H1), statuses + formula vs direct regression (H2), robustness to wording (H3).

Overall quality and quality of each step

Data generation	Generator fidelity: verifier and human agreement with the requested status
Classification	Macro-F1 over D/P/O/C; per-class F1
End-to-end score	Spearman, MAE (score); QWK (categories)
Safety	Dangerous-miss and omission-miss rates + Wilson 95% CI; over-flag rate on fully correct restatements
Robustness	Score gap and category flips on meaning-matched pairs

Protocols and ground truth

- Ground truth: verified synthetic labels; challenge set adjudicated by two annotators (weighted κ reported).
- Three test sets reported separately: familiar instructions, held-out drug classes, human-written.
- 3 seeds per model, mean $\pm$ std.
- Sensitivity: rerun with $\pm$50% weights and partial = 0.25; check rankings hold.

Dangerous-miss = answers with a critical **C** placed in "no flagged" $\div$ all answers with a critical **C**. Omission-miss: the same for a critical **O**/**P**. 0 / 300 $\Rightarrow$ 95% upper bound ≈1.3%.

Backup

Backup: worked example

Instruction: "Take one 500 mg amoxicillin capsule three times a day for 7 days. Finish all of it even if you feel better. Call your doctor right away if you get a rash, swelling or trouble breathing."

Restatement: "I take one capsule three times a day, and I can stop once I feel better."

Component	Category	Critical	$\tilde{w}_i$	Status	Points
One 500 mg capsule per dose	dose	yes	0.357	**D** ("one capsule")	$+0.357$
Three times a day	timing	no	0.286	**D**	$+0.286$
For 7 days	duration	no	0.071	**C** ("stop once I feel better")	-0.036
Finish even if better	duration	no	0.071	**C**	-0.036
Rash / swelling / breathing $\rightarrow$ call doctor	warning	yes	0.214	**O**	0

$S = 57 \Rightarrow$ **clarification recommended**. **Must check**: allergy warning signs (critical, omitted; the 84 cap holds but 57 is already lower). Revisit: course length.

If "finish the course" were flagged critical: $\min(57, 40) = 40 \Rightarrow$ substantial misunderstanding. Dose wrong but all else right, in a 7-category instruction: 62.5 without the cap $\Rightarrow$ capped to 40. Only "no alcohol" (weight 0.10, critical) omitted, all else right: 90 $\Rightarrow$ capped to 84, clarification recommended.

Backup: scoring rule details and scope

Category base weights

dose	0.25	warning signs	0.15
timing	0.20	duration	0.10
interactions	0.20	administration	0.05
		missed dose	0.05

Split equally among a category's components, then renormalised over the categories present. Credit: $D = 1$, $P = 0.5$, $O = 0$, $C = -0.5$ (a wrong belief is worse than none).

Safety caps: critical **C** $\Rightarrow S \leq 40$; critical **O**/**P** $\Rightarrow S \leq 84$. "No flagged" requires every critical component to be **D**. Omission = missing evidence, not proven error, hence the softer cap.

Weights, credits, caps and thresholds are *hypotheses*, reviewed by a pharmacist or nurse if available, otherwise based on label text and the ISMP high-alert list.

Scope

- Measures *demonstrated* understanding in the restatement, not what the patient knows.
- Components are given (structured instruction); automatic decomposition, speech and corrective advice are out of scope.
- Decision support with a clinician in the loop; synthetic and role-played data only, no patient records.

Closest work

- Teach-back studies: recall scored by hand.
- DischargeSim (2025): multiple-choice comprehension exams.
- HealthLit (CHASE 2025): scores the clinician's practice.
- Short-answer grading: no safety weighting.